

Compiled Some Data on AI energy usage VS Netflix Streaming

Posted by u/UniqueNameTaken • 0 points • 16 comments

Got bored one night, and I decided to look this up since I saw so much back and forth on how much energy AI uses. So, MIT notes that the average amount of energy expended to create an "average" level text response is 114 Joules of energy (essentially the amount of energy used to power a microwave 1/10 of a second). The beefiest monster of an LLM takes 6706 Joules of energy. This accounts for the energy needed to run the calculations, along with other systems like cooling and transmissions.

Image generators are beefier per request at 2282 Joules for an average image. Higher quality obviously makes that number go up.

So now we have data expenditure for LLMs and image generation. Let's compare that to watching Netflix, because someone did a study on that in 2019. This is from the International Energy Agency, which is a Paris-based, non-governmental agency that studies data.

This study showed that the closest average estimate they can make (data centers and streaming services don't usually like to share exact data) is per hour of Netflix watching .077 kWh of energy is spent. Based on their charts, we can see the average amount of energy being used there by the data centers and transmission is about 28% of the total energy (the rest of that going toward powering your device while watching Netflix). So let's adjust the per hour: $.077 \times .28 = .02156$ kWh. So let's convert that to Joules: 77616 Joules/hour of streaming. Let's make the numbers a bit easier and comparable and divide by 60 for a per-minute: 1293.6 Joules per minute of streaming.

Final Counts:

114 Joules per average text response

6706 Joules per maximum level text response

2282 Joules per average image generation

77616 Joules per hour of Netflix

1294 Joules per minute of Netflix

<https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-footprint-big-tech/>

<https://www.iea.org/commentaries/the-carbon-footprint-of-streaming-video-fact-checking-the-headlines>

Comments

u/god_oh_war • 1 points • 2025-07-10 08:46 UTC

The environmental argument has already been so thoroughly defeated that it's insane people still use it.

u/Kyokyodoka • 0 points • 2025-07-10 08:53 UTC

And how can we prove it?

Frankly at this point it seems more likely they are lying about the amount of pollution / water waste given its not unlikely they are hiding the failings of AI under a heavy blanket.

u/god_oh_war • 1 points • 2025-07-10 08:56 UTC

I mean, you can run a model locally and compare it to other activities like gaming or rendering something on Blender :P

If you own a computer, that thing is using electricity, and if someone decides they want to generate an image rather than play a videogame, both are still using very similar amounts of electricity.

(And ofc because both use electricity it directly translates into both wasting the same amount of water.)

u/Purplescheme • 1 points • 2026-01-17 03:51 UTC

It's not only about the power consumption at any given moment, but rather the overall increase in power needed to power these behemoths for the rest of time (?). Powergrids across the country are being treated like an abused mother and we (the ratepayers) are the red-headed stepchild.

u/im_eddie_snowden • 2 points • 2025-07-10 01:00 UTC

You can practically cook an egg on a MacBook Pro running Photoshop for digital art.

u/god_oh_war • 1 points • 2025-07-10 08:46 UTC

Okay but why would you use Mac 🤔

u/im_eddie_snowden • 1 points • 2025-07-10 11:08 UTC

I usually don't, but a lot of graphic designers do.

u/fathersmuck • 1 points • 2025-07-09 23:49 UTC

The closed AI model providers are serving up a total black box," says Boris Gamazaychikov, head of AI sustainability at Salesforce, who has led efforts with researchers at Hugging Face, an AI platform provider of tools, models, and libraries for individuals and companies, to make AI's energy demands

more transparent. Without more disclosure from companies, it's not just that we don't have good estimates—we have little to go on at all.

u/SoberSeahorse • 2 points • 2025-07-10 02:06 UTC

Boris who? Sounds made up.

u/fathersmuck • 0 points • 2025-07-10 02:50 UTC

It is part of the article he used to figure out AI power usage.

u/Tyler_Zoro • 8 points • 2025-07-09 22:36 UTC

PLEASE stop citing that tech review article! It's utter crap. They admit that they assume the use of H100 GPUs dedicated to a single request. For you to then try to generalize that to any AI image generation is absolutely absurd.

You want to come over and measure the power draw from my PC to generate a single image, you feel free. I guarantee it will make that article look very silly (hint: I literally ***can't** draw that kind of power on the circuit my PC is connected to).

u/malibusoul • 1 points • 2025-12-29 01:26 UTC

I'm just merely curious - are you PRO or ANTI AI? I've looked through your comments and I actually can't tell, lol.

u/WeirdExcuseMonkey • 0 points • 2025-11-23 23:33 UTC

Wrong

u/UniqueNameTaken • 2 points • 2025-07-10 00:27 UTC

I'm happy to update my data if there is another reliable article out there. As noted by both the links I cited, data centers, streaming services, AI systems, and I'm sure many other industries don't like sharing their power usage rates, so there is a lot of guesstimation involved. Both MIT Tech Review and IEA have articles where they admit they are estimating and are doing their best to correct for variables. However, both are considered trusted sources with high reliability in their reporting. I think if nothing else, this gives a good rough look at energy usage in a head-to-head. If someone else finds a better article, I'll adjust the data around to match.

u/Adoreball • 6 points • 2025-07-09 23:02 UTC

I think it works in this context. If you wildly, intentionally overestimate and still come up with a number too small to worry about, then the point has been proven.

u/Tyler_Zoro • 5 points • 2025-07-09 23:18 UTC

It's still not a useful source to cite. That's the only point I was making.

u/SoberSeahorse • 6 points • 2025-07-09 22:03 UTC

So antis that watch Netflix can eat my entire ass?

u/axiaelements • 3 points • 2025-07-10 01:38 UTC

Please practice proper hygiene.

u/Mikhael_Love • 5 points • 2025-07-09 22:36 UTC

Just the inside.